

AEQUITAS NODE OPERATOR GUIDE

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Complete step-by-step guide · No prior blockchain experience needed · ~20–30 min

Aequitas Chain · Chain ID 1926 · EVM Compatible · Go 1.24

What is an Aequitas Node?

An Aequitas node runs in the cloud, validates human registrations, produces blocks, and keeps the blockchain alive. Node operators earn a daily share of protocol fees — automatically every day at **20:00 Berlin time (CEST/CET)**. No minimum uptime required.

■ **IMPORTANT:** To receive validator rewards, **NODE_OPERATOR_WALLET** must be your registered Aequitas human wallet — the one verified with AequitasBio. Only verified humans earn validator rewards.

Before You Start

- 1 **Aequitas account:** Register via the AequitasBio Android app. You need your human wallet address to receive rewards. If not yet registered, do that first.
- 2 **GitHub account (free):** Create at github.com — you need it to fork the Aequitas repository.
- 3 **Railway account (free):** Sign up at railway.app using GitHub. No server or command line needed.
- 4 **Signing wallet (MetaMask):** Create a **separate** MetaMask wallet for your node to sign transactions. Export its private key: MetaMask → Account Details → Show Private Key. Keep it strictly private.
- 5 **10–30 minutes:** Railway automates the build and deployment.

Environment Variables

■ **RELAYER_PRIVATE_KEY** is a private key. Never share it, never paste it in chat or email. Use a dedicated wallet — not your personal human wallet.

Variable	Required?	Description
<code>DATABASE_URL</code>	YES	PostgreSQL connection string. Railway auto-injects when PostgreSQL is in the same project.
<code>RELAYER_PRIVATE_KEY</code>	YES	Private key of your signing wallet (0x..., 66 chars). MetaMask → Account Details → Show Private Key.
<code>RELAYER_ADDRESS</code>	Recommended	Public address matching RELAYER_PRIVATE_KEY (0x..., 42 chars). Prevents startup errors.

Variable	Required?	Description
NODE_OPERATOR_WALLET	For rewards	Your registered HUMAN wallet on Aequis. Receives daily validator rewards at 20:00 Berlin. MUST be a verified human.
PEER_SECRET	Multi-node	Shared secret — all nodes in the network must use the SAME value. Obtain from the network operator. Do not publish.
SELF_URL	Multi-node	Your node's public URL: https://YOUR-NAME.up.railway.app
PRIMARY_NODE_URL	Multi-node	Set to: https://aequis.digital — your node auto-registers here on startup.
NODE_KEY	Optional	Base64 libp2p key for stable peer ID. If not set: auto-generated and printed to stderr as "SAVE THIS AS NODE_KEY: ". Copy and set it.
IS_PRIMARY_NODE	NO	Leave unset. Distribution now uses a DB-level lock — any node can run it without this variable.
RESET_STATE	NO	DANGEROUS: wipes the entire database on restart. Development only. Never in production.

Step-by-Step Deployment (Railway)

1 Fork the Repository

Open github.com/hanoi96international-gif/Aequis in your browser. Click **Fork** in the top-right corner, then **Create fork**. GitHub creates a copy under your account.

2 Create PostgreSQL Database

Go to railway.app, sign in with GitHub. Click **New Project** → **+ New** → **Database** → **Add PostgreSQL**. Railway creates the database and will auto-inject DATABASE_URL.

3 Deploy Your Node

In the same Railway project: **+ New** → **GitHub Repo** → select your Aequis fork. Railway detects the Dockerfile automatically. Click **Deploy Now**.

4 Set Environment Variables

Click your Aequis service → **Variables**. Add the required variables (see table above). Minimum required:

- RELAYER_PRIVATE_KEY
- RELAYER_ADDRESS
- NODE_OPERATOR_WALLET
- PEER_SECRET
- SELF_URL
- PRIMARY_NODE_URL = https://aequis.digital

Save — Railway auto-redeploys.

5 Get Your Public URL

Settings → Networking → **Generate Domain**. Open <https://YOUR-URL/api/status> in your browser. You should see JSON with "height" climbing every ~6 seconds.

6 Verify in Logs

Check Deploy Logs for:

- ✓ API Server listening on port 8080
- ✓ [SYNC] Connected to peer <https://aequitas.digital>
- ✓ [NODE] Registered node operator wallet: 0x...

7 Earning Rewards

Rewards are distributed automatically every day at **20:00 Berlin time (CEST/CET)**. The Validators Pool (40% of all protocol fees: swap fees, demurrage, wealth cap overflow) is split proportionally among all registered node operators by blocks produced. Keep your node running — no manual action needed.

Alternative: Docker Deployment

```
git clone https://github.com/hanoi96international-gif/Aequitas && cd Aequitas
docker build -t aequitas-node .
docker run -d --name aequitas --restart unless-stopped \
  -e DATABASE_URL="postgres://user:pass@host:5432/aequitas" \
  -e RELAYER_PRIVATE_KEY="0xYOUR_KEY" \
  -e RELAYER_ADDRESS="0xYOUR_ADDRESS" \
  -e NODE_OPERATOR_WALLET="0xYOUR_HUMAN_WALLET" \
  -e PEER_SECRET="from-network-operator" \
  -e SELF_URL="https://YOUR-URL" \
  -e PRIMARY_NODE_URL="https://aequitas.digital" \
  -p 8080:8080 aequitas-node
```

Troubleshooting

Symptom	Cause	Solution
Height stays at 0	PRIMARY_NODE_URL missing	Set PRIMARY_NODE_URL= https://aequitas.digital and also set SELF_URL. Redeploy.
DATABASE_URL error	Wrong connection string	Format: postgres://user:pass@host:5432/dbname
"no code at address"	V7 not deployed yet	Normal on first start — node deploys V7 automatically. Wait 10–15 seconds.
No rewards received	NODE_OPERATOR_WALLET wrong	Must be your registered HUMAN wallet on Aequitas (verified with AequitasBio).
"Application error" Rail.	Build or startup failed	Check Deploy Logs. Common causes: missing DATABASE_URL or wrong key format.
Port 8080 unreachable	Firewall config (Docker)	Open TCP port 8080 inbound in your firewall or cloud security group.